Project Design Phase-II
Technology Stack (Architecture & Stack)

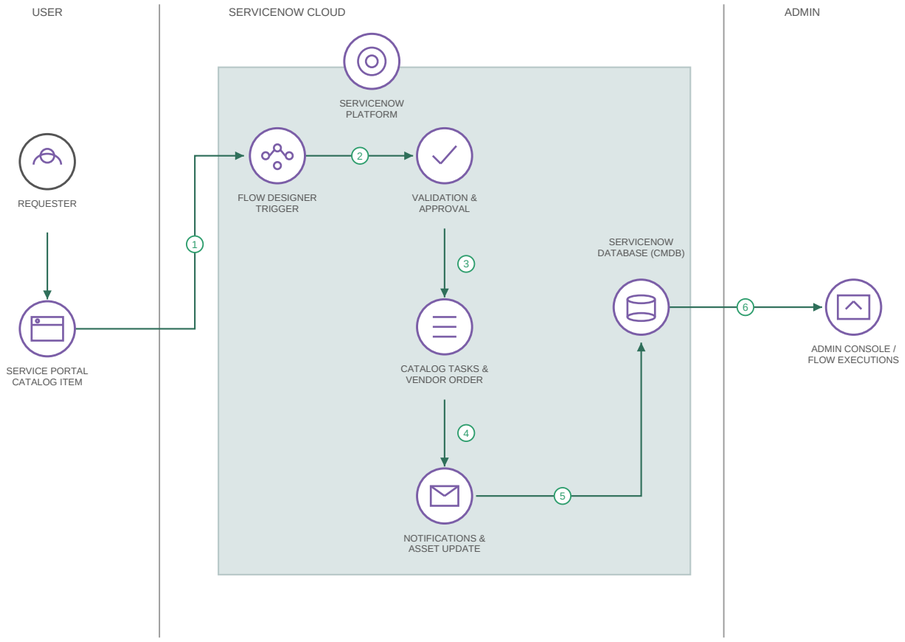
Date	31 January 2025
Team ID	SWTID-2026-1951
Team Members	Nancy.D, Mythely.J, Srimathi.K, Sahasri.C
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Standard laptop order automation with ServiceNow Flow Designer

Reference: https://docs.servicenow.com/



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How user interacts with application e.g. Service Portal, Mobile App	ServiceNow Service Portal (HTML, CSS, JavaScript / AngularJS), ServiceNow Mobile App
2.	Application Logic-1	Logic for request intake in the application	Catalog Item, Variables & Client Scripts (JavaScript)
3.	Application Logic-2	Logic for validation and approval automation	ServiceNow Flow Designer (Flow, Subflow, Approval action)
4.	Application Logic-3	Logic for fulfillment and notifications	Flow Designer actions (Create Catalog Task, Send Email)
5.	Database	Data Type, Configurations etc.	ServiceNow tables (sc_request, sc_req_item, sc_task)
6.	Cloud Database	Database Service on Cloud	ServiceNow platform database, CMDB
7.	File Storage	File storage requirements	ServiceNow Attachments (sys_attachment)
8.	External API-1	Purpose of External API: vendor catalog, stock check, and order placement	Vendor Procurement REST API (REST Message / IntegrationHub)
9.	External API-2	Purpose of External API: sending notifications to users	Email / SMTP notification service
10.	Machine Learning Model	Purpose of Machine Learning Model: classify and route requests (optional)	ServiceNow Predictive Intelligence (optional)
11.	Infrastructure (Server / Cloud)	Application Deployment on Local System / Cloud Local Server Configuration: Cloud Server Configuration :	ServiceNow cloud-hosted instance Not required (cloud-hosted) ServiceNow instance with Service Catalog and Flow Designer active

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used	AngularJS (Service Portal widgets), Bootstrap
2.	Security Implementations	List all the security / access controls implemented, use of firewalls etc.	Role-based access control (ACLs), SSO, HTTPS / TLS encryption
3.	Scalable Architecture	Justify the scalability of architecture (multi-tier cloud platform, reusable subflows)	ServiceNow multi-tier cloud platform, Flow Designer subflows
4.	Availability	Justify the availability of application (e.g. cloud redundancy, managed platform)	ServiceNow cloud high availability and data center redundancy
5.	Performance	Design consideration for the performance of the application (asynchronous flow execution, indexed tables)	Asynchronous flow execution, table indexing, caching

References:

https://c4model.com/

https://docs.servicenow.com/

https://www.ibm.com/cloud/architecture

https://aws.amazon.com/architecture

https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d